

Overfitting and How Random Forest Solves It

Overfitting is a common problem in machine learning where a model learns the training data too well, including its noise and small irregular patterns. As a result, the model performs extremely well on the training dataset but fails to make accurate predictions on new, unseen data. In simple terms, the model memorizes the training examples instead of learning the general patterns that can be applied to future data.

A Decision Tree is particularly vulnerable to overfitting when it is allowed to grow very deep. The tree keeps creating branches to correctly classify every training sample. This leads to very high training accuracy, sometimes reaching 100%, but the model may perform poorly on the test dataset because it has become too specific to the training data.

Random Forest is an ensemble learning algorithm designed to reduce overfitting. Instead of relying on a single Decision Tree, it creates many Decision Trees using different random subsets of the training data and features. Each tree makes its own prediction, and the final prediction is determined through majority voting.

Because every tree sees a slightly different version of the data, individual mistakes and overfitting tendencies are reduced. The collective decision of multiple trees is usually more accurate and stable than the prediction of a single tree. Random Forest therefore improves generalization, increases robustness, and provides better performance on unseen data.

In this project, the Random Forest model achieved higher and more stable accuracy compared to the Decision Tree model. The use of multiple trees helped reduce overfitting and produced more reliable predictions for classifying Iris flower species.